

AI-driven sustainable agriculture in reducing environmental impact: the role of market-oriented strategies

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Abstract

Purpose – This study aims to examine how artificial intelligence (AI) driven sustainable agriculture practices contribute to environmental sustainability among farmers in Ghana. It also examines the mediating role of market-oriented strategies in this relationship.

Design/methodology/approach – The research used a quantitative approach and used structural equation modeling to investigate the impact of AI adoption on reducing environmental impact, mediated by market orientation strategies. The data were collected from 211 individual farmers who actively use/adopted an AI in their farming.

Findings – The study identifies that the adoption of AI technologies has a strong and significant impact on improving crop management, resource efficiency and reducing carbon footprint. However, the mediating effect of market-oriented strategies showed mixed results for access to market and price mechanisms.

Research limitations/implications – While the study offers a valuable conceptual framework, its generalizability may be improved with a bigger sample. Future studies are encouraged to conduct the study in other jurisdictions to validate the proposed relationships.

Originality/value – This study contributes to the literature by bridging the gap between AI technological innovations and market strategy within the sustainability discourse. This paper goes beyond the techno-centric approach to reducing environmental impact and presents a holistic model that emphasizes market-oriented strategies as a key enabler of Adopting AI in reducing environmental impact within the agriculture sector.

Keywords Artificial intelligence, Sustainable agriculture, Market-oriented strategies, Environmental impact, SME farmers, Ghana

Paper type Research paper

1. Introduction

Agriculture remains a cornerstone of Ghana's economy, underpinning livelihoods and contributing substantially to the national gross domestic product (GDP). In 2023, the agricultural sector accounted for 21.1% of Ghana's GDP, reflecting its significant role in the country's economic framework (World Bank, 2023). Despite its centrality, the sector is beset with persistent structural inefficiencies, particularly among small and medium-sized enterprise (SME) farmers. Traditional agricultural practices are characterized by suboptimal resource utilization, environmental degradation, low productivity and constrained access to markets, factors that collectively undermine both sustainable development and economic resilience (Yang and Solangi, 2024). These challenges are exacerbated by the intensifying pressures of climate change, land degradation and demographic expansion, which necessitate



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innovative approaches to ensure environmental sustainability and agricultural competitiveness. Recent technological advancements, particularly those driven by artificial intelligence (AI), present significant opportunities for transforming agricultural practices. Globally, AI-enabled technologies such as precision agriculture, intelligent irrigation, crop monitoring systems and predictive analytics have demonstrated considerable potential in enhancing agricultural productivity while mitigating environmental externalities (Singh *et al.*, 2022; Javaid *et al.*, 2023; Eli-Chukwu, 2019). These innovations facilitate data-driven decision-making, optimize input utilization, reduce wastage and enhance the efficiency of farming operations. Nevertheless, the diffusion and application of such technologies within the Ghanaian agricultural landscape, especially among SME farmers, remains limited and fragmented (Balkrishna *et al.*, 2023). The barriers to adoption are multifaceted, encompassing prohibitive costs, limited awareness, inadequate digital infrastructure and low levels of technological literacy.

While extant literature has extensively explored AI technological capabilities such as automation and greenhouse efficiency to the broader promise of future framing (Kumari *et al.*, 2025; Altayeb *et al.*, 2024; Hoseinzadeh and Garcia, 2024; Javaid *et al.*, 2023; Ryan *et al.*, 2023; Issa *et al.*, 2022; Jha *et al.*, 2019), limited academic attention has been given to the strategic mechanisms that enable or constrain the effective deployment and utilization of these technologies and the pathways from technological adoption to tangible environmental outcomes in real-world agricultural contexts. In particular, market-oriented strategies such as access to markets and pricing mechanisms (De Toni *et al.*, 2022) offer a vital pathway for contextualizing and operationalizing AI-driven innovations within local agricultural markets. These strategies serve not merely as ancillary considerations but as pivotal enablers that align technological solutions with market needs, enhance value delivery and promote sustainable competitive advantage. Accordingly, the interaction between AI-driven agricultural practices and market-oriented strategies warrants more rigorous scholarly attention. To address this gap, this study examines the impact of AI-driven sustainable agricultural practices on environmental sustainability within the SME agricultural sector in Ghana, with a particular focus on the mediating role of market-oriented strategies. By positioning market-oriented strategies as a central mediating construct, the study advances the understanding of how technological innovations and strategic market practices coalesce to enhance environmental outcomes in agriculture. This integrative approach moves beyond predominantly techno-centric perspectives by offering a more holistic framework that incorporates both technological innovation and strategic market engagement. The study provides empirically grounded insights that are directly relevant to policymakers, agricultural practitioners, development organizations and SME farmers. By elucidating pathways through which AI technologies, when effectively coupled with robust market strategies, can mitigate environmental degradation and enhance agricultural resilience, the study contributes substantively to national and global sustainable development agendas. In doing so, the study aligns with Sustainable Development Goal (SDG) 2 which stress on food security and sustainable food production systems, (Zero Hunger), SDG 8 (Decent Work and Economic Growth), which focus on the benefits of inclusive economic participation in agriculture value chains and SDG 13 (Climate Action), which highlights the mitigation and adaptive strategies address climate change effects underscoring the interlinkages between technological innovation, market orientation, environmental sustainability and sustainable agriculture within emerging economy contexts.

2. Literature review

2.1 The triple bottom line

This study is guided by the TBL. The TBL was put forth by Elkington (1994) to broaden the environmental sustainability agenda by incorporating social. It is a comprehensive

framework that evaluates firm performance by integrating economic, environmental and social dimensions from a sustainability perspective (Elkington, 1997). The framework can lead organizations to achieve economic prosperity, environmental quality and social justice simultaneously (Gu et al., 2021). Within the study, the framework assesses how AI-driven technologies and market-oriented strategies can contribute to minimizing environmental effects while maintaining economic viability and social responsibility.

From an environmental perspective, the framework highlights how AI-driven technologies in agriculture can minimize negative environmental effects by enhancing resource efficiency. Tools such as smart irrigation systems, precision farming and predictive analytics can reduce water usage, lower greenhouse gas emissions and reduce chemical inputs. These innovative ideas foster sustainable farming practices. Economically, the framework highlights economic viability with sustainability. AI-driven agricultural technologies can enhance market access, minimize cost and increase yields by ensuring sustainability efforts remain profitable (Padhiary et al., 2025). Socially, AI-driven agricultural technologies can improve decision-making, productivity and strengthen food security. Nevertheless, it also raises concerns about unequal access to AI-driven technologies, which can be mitigated by market-oriented strategies that can promote inclusive participation and equitable value chains.

2.1.1 Artificial intelligence adoption. Arguably, AI is the oldest domain within the field of computer science (Holzinger et al., 2019) and emerged as a fundamental component of the technology industry, providing significant solutions for informatics challenges (Abdussalam and Auladi, 2024). The term was coined by computer scientist John McCarthy in 1979. He defined it as “the science and engineering of making intelligent machines” (Herath and Mittal, 2022, p. 2). The OECD (2016) and UNCTAD (2017) consider AI as the ability of machine systems to obtain and used knowledge to undertake intelligent behaviors. It focuses on developing programs that can perform tasks that typically require human thinking or intelligence (Ghosh and Thirugnanam, 2021; Waly, 2024). This includes a range of cognitive tasks such as decision-making, learning, reasoning, processing oral languages and sensing, and also moving and manipulating accordingly (Ren et al., 2024). These intelligent systems include a blend of cloud computing, machine-to-machine communication, big data analytics and the internet (Sinha and Gond, 2024). Some of the objectives of AI are replication of human cognitive abilities, developing machines to perform tasks that require human abilities and developing systems that can autonomously learn from their experiences (Ghosh and Thirugnanam, 2021).

Reports from McKinsey & Company (2019) revealed that AI adoption has been on the rise, leading to a significant growth in revenue and reduction of cost, particularly among AI adopters across different operational activities. Furthermore, Singh (2025) argued that AI has been extensively applied in many fields, with its focus on the technology domain, and has created 800 million jobs by 2030. Literature indicates that this cutting-edge technology has revolutionized nearly every sector, including manufacturing (Li et al., 2025), engineering (Puttegowda and Nagaraju, 2025), medicine (Holzinger et al., 2019), education (Ayeeni et al., 2024) and, more importantly, the agricultural sector.

2.1.2 Artificial intelligence-driven sustainable agriculture. To meet and satisfy the needs of the current generation without compromising that of future generations, it is important to use sustainable strategies to manage land and natural resources (Jha and Sharma, 2025; Tilavova et al., 2024). This, according to Padhiary et al. (2025), involves the integration of environmental, economic and social goals to ensure ecological sustainability, productivity and community well-being. Conventional farming techniques serve as a challenge to agricultural practitioners (Sekhar et al., 2024). Historically, the agricultural sector was primarily focused on the cultivation and production of crops and food (Porter et al., 2019). However, the last two

decades have witnessed a substantial transformation that broadened its scope beyond conventional practices (Pretty, 2018). Currently, agricultural practices involve the production of crops and animal products, to subsequent distribution, marketing and processing of these products on an industrial scale (Raihan, 2024). At present, the sector is the source of food, contributes to GDPs, fosters national and international trade, creates employment opportunities, supplies raw materials for other industries and promotes economic stability and growth (Kumar, 2024). As a result, agricultural techniques must be evaluated to offer innovative solutions for a long-term impact and improve productivity, considering the rapid growth of global population (Muchhadiya *et al.*, 2024; Muhie, 2022). Artificial emerged as a revolutionary agricultural technique that mitigates these challenges (Raihan, 2024; Szymański *et al.*, 2025).

The adoption of AI in agricultural practices enables farmers to access information about soil and pest management, optimal crop yield and improves many tasks throughout the food supply chain, thus assisting agricultural practitioners (Javaid *et al.*, 2023). These scholars (Aggarwal *et al.*, 2024) observed that AI in the context of agricultural practices includes different applications that relate to precision farming, predictive analytics, intelligent advisory systems, crop health monitoring, agricultural robotics and predictive analytics. These are the specific areas where agrarian practices hold the innovation potential, ranging from climate unpredictability, resource inefficiency and labour shortages (Kamilaris and Prenafeta-Boldú, 2018). These technologies have the potential to enhance soil health monitoring, crop-yield forecasting, as well as irrigation and weeding (Dawn *et al.*, 2023).

2.1.3 Research hypotheses. Literature acknowledges that the integration of AI into agricultural practices has led to significant improvements in the sector, particularly in environmental factors such as resource efficiency, crop management and reduction of carbon footprints. For instance, Ahuja and Mehra (2023) noted that this cutting-edge technology has enhanced the productivity and efficiency of India's agricultural sector. Adewusi *et al.* (2024) noted that AI not only boosts productivity but also alleviates issues related to labor shortages, thereby facilitating a more economically and sustainably viable future for the agricultural sector. Similarly, Sutton *et al.* (2020) observed that one strength of machine learning in decision support systems is its ability to predict future trends and outcomes. This, according to Hu *et al.* (2024) is accomplished by analyzing weather patterns, historical data and crop-specific parameters. ML algorithms can forecast crop yields, determine optimal planting periods and anticipate potential issues such as disease outbreaks. AI-powered irrigation systems also use real-time data from various sources, such as weather forecasts, soil moisture and crop management, to accurately control the timing and volume of irrigation (Sinwar *et al.*, 2019), thereby leading to improved efficiency and productivity (Abokyi *et al.*, 2020).

Regarding crop management, Hamed *et al.* (2024) mentioned that precision agriculture, powered by AI, aims to enhance crop management through real-time data analysis. As demonstrated by Borah *et al.* (2025) and Omia *et al.* (2023), AI-powered drones and sensors for planters enable the monitoring of soil conditions, environmental factors and crop health and strain. This comprehensive data facilitates customized techniques, such as fertilization and targeted irrigation, leading to increased crop yield and minimizing resource waste (Al-Atrash *et al.*, 2020). Ahuja and Mehra (2023) noted that AI can predict types of crops suitable for planting in specific areas and years, as well as the optimal harvesting dates. Furthermore, AI-driven robotics and autonomous vehicles have primarily been used for weeding, planting and harvesting, thereby reducing reliance on manual labor while enhancing efficiency in agricultural operations (Altayeb *et al.*, 2024).

Accordingly, Fasciolo *et al.* (2024) reported that the agricultural sector is responsible for approximately 70% of global freshwater usage while contributing to greenhouse gas emissions. As demonstrated by Elshaikh-Hayat *et al.* (2024), the integration of AI-powered

irrigation systems can reduce water consumption by 30% through precise soil moisture monitoring and predictive analytics. [Rajak et al. \(2023\)](#) also highlighted that AI tools, such as deep learning algorithms and IoT-based sensor networks, can decrease agricultural carbon emissions by 35% by optimizing resource use. This is supported by [Ahuja and Mehra \(2023\)](#), who argued that sustainable AI tools can lower greenhouse gas emissions by 25%, reduce water consumption by 40% and enhance productivity by 30%. [Saran \(2019\)](#) reported that by 2030, the application of AI could minimize greenhouse gas emissions from 2.7% to 4.8% in East Asia, and from 1.6% to 6.1% in North America. Based on these, we hypothesized that:

- H1. There is a positive relationship between AI adoption and resource efficiency.
- H2. There is a positive relationship between AI adoption and crop management.
- H3. There is a positive relationship between AI adoption and a reduction in crop management.

2.1.4 Market-oriented strategies. Market orientation refers to an organizational culture that focuses on generating market intelligence about current and future customer needs, disseminating information to all departments and collectively responding to this information ([Kohli and Jaworski, 1990](#); [Mahmoud et al., 2019](#); [Narver and Slater, 1990](#)). This concept has been studied within the manufacturing ([Kanaan-Jebna et al., 2022](#)), banking ([Mahmoud et al., 2016](#)) and tourism ([Tsetse et al., 2022](#)), and it is described as the extent to which a manager leverages market knowledge, particularly pricing and customers, as a basis for decision-making on the three basic questions in economics, i.e. what and how to produce and how to market ([Gebremedhin, 2010](#)). In context, it pertains to production-related decisions influenced by market signals and production conditions ([Otekunrin et al., 2019](#)). [Amadasun and Mutezo \(2022\)](#) observed that market-oriented strategies center on delivering superior value to customers, which shapes firms' strategic direction to achieve market growth. In this context, literature ([Ilak et al., 2021](#); [Zhang et al., 2021](#)) identified access to markets and pricing mechanisms as some of the market-oriented strategies used by small-scale farmers. Past studies have examined these strategies and their significance to small-scale farmers. Interestingly, the majority of the prior studies measured the market-oriented strategies in fragments, concentrating on pricing mechanisms and access to markets in isolation, without examining these strategies holistically. This minimizes the comprehension of how the integration of these market-oriented strategies by small-scale farmers helps in improving environmental outcomes.

The relationship between artificial intelligence-driven agricultural practices and environmental impacts, such as resource efficiency, crop management and reduction in carbon footprint, may be influenced by market-oriented strategies such as access to market and pricing mechanisms. Small-scale farmers who use AI in their practices but cannot access markets may find it challenging to effectively implement sustainable practices ([Magesa et al., 2020](#)). As indicated by [Agholor et al. \(2023\)](#), access to markets enables farmers to sell their produce at competitive prices. [Otekunrin et al. \(2019\)](#) also observed that access to markets by small-scale farmers makes significant contributions to the development and growth of agriculture in rural areas, thus improving productivity. Similarly, research indicates when pricing mechanisms such as incentives ([Nindi et al., 2024](#)), direct sales ([Corsi et al., 2018](#)), price support through buffer stock ([Abokyi et al., 2020](#)), cooperative model ([Wang et al., 2022](#)), price support programs ([Abokyi et al., 2020](#)), as well as forward contracts ([Meemken and Bellemare, 2020](#)), among others, are consistent with farmers sustainable objectives, small scales farmers are willing to use innovative techniques that will improve productivity ([Paudel et al., 2019](#)). Price is recognized as

an important element for farmers (Naruetharadhol *et al.*, 2022). Thus, the ability to set effective and efficient prices represents intangible resources for small-scale farmers to improve their efficiency (Naruetharadhol *et al.*, 2022). Despite these studies, there appears to be a paucity of studies mediating market market-oriented strategies, such as access to markets and pricing mechanisms, between AI-driven sustainable agricultural practices and environmental impact. Figure 1 depicts the conceptualization of the study and we therefore hypothesize that:

- H4a. Access to markets mediates the relationship between AI-driven sustainable agricultural practices and resource efficiency.
- H4b. Access to markets mediates the relationship between AI-driven sustainable agricultural practices and crop management.
- H4c. Access to markets mediates the relationship between AI-driven sustainable agricultural practices and a reduction in carbon footprint.
- H5a. Pricing mechanisms mediate the relationship between AI-driven sustainable agricultural practices and resource efficiency.
- H5b. Pricing mechanisms mediate the relationship between AI-driven sustainable agricultural practices and crop management.
- H5c. Pricing mechanisms mediate the relationship between AI-driven sustainable agricultural practices and a reduction in carbon footprint.

3. Methodology

The present study was conducted in Ghana, an emerging economy where agriculture plays a pivotal role in national development and the sustenance of livelihoods (Azumah *et al.*, 2023). Ghana relies heavily on agriculture for employment opportunities, food security and income generation, particularly in the rural areas of the country (Koryo-Dabrah *et al.*, 2021). The sector is also characterized by SME farms, which are the pillars of agricultural production, but often challenged with technology adoption, productivity and market access (Aidoo *et al.*, 2025; Quaye *et al.*, 2024). The research specifically targeted small and medium-sized enterprises (SMEs) farmers and agricultural crop producers within two prominent agricultural regions: Greater Accra and the Eastern Region (Afful-Koomson and Fonta, 2015; Essel *et al.*, 2026). These regions were purposively selected due to their distinct

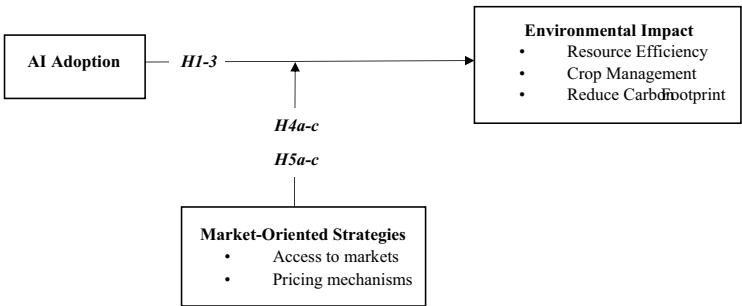


Figure 1. Conceptual framework
Source: Developed by authors, 2025

agricultural landscapes which are broadly representative of the diverse farming systems prevalent across Ghana and active internet connectivity that ensures farmers' interactions with AI-driven technologies. Moreover, the choice of these areas was informed by the farmers' deep familiarity with the land, crops and local agricultural systems, positioning them as key actors capable of influencing and advancing improved farming practices. In collaboration with community leaders and local agricultural associations, contact details including phone numbers and farm locations of potential study participants were obtained.

A purposive sampling technique (Thomas, 2022) was used to select SME farmers and agricultural crop producers operating within the two regions. The inclusion criteria stipulated that participants must have been engaged in farming activities for a minimum of two years and actively using AI technologies in their agricultural practices (Keung *et al.*, 2020). These criteria ensured the selection of participants with sufficient experience and familiarity with AI applications in farming. A total of 235 SME farmers and agricultural crop producers were approached through phone calls and in-person visits to their farms to invite their participation in the study. Of this number, 8 were excluded due to non-usage of AI in their farming operations, 5 were excluded for having less than two years of farming experience and 6 declined participation citing reasons such as time constraints, unavailability or personal circumstances. Following additional follow-ups, 216 farmers and agricultural crop producers consented to participate in the study. These participants represented a diverse range of farming backgrounds across the selected regions. All respondents met the inclusion criteria, possessing at least two years of farming experience and actively using AI-driven innovations to enhance agricultural productivity. Data collection was conducted using a semi-structured questionnaire. Informed consent was obtained from all participants, and ethical guidelines were rigorously adhered to throughout the research process (Xu *et al.*, 2020). The questionnaire was structured into two main sections. Section 1 focused on assessing the key constructs central to the study, namely, AI adoption, environmental impact and market-oriented strategies. Section 2 collected demographic information from the respondents. Responses to the questionnaire were recorded using a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). The measurement scales for each construct were adapted from validated instruments used in prior scholarly studies to ensure both reliability and validity. The researchers conducted a pretest before the main data collection with 12 small and medium-sized enterprise (SME) farmers and agricultural crop producers from the two selected regions to assess the clarity of the questionnaire, eliminate ambiguity and ensure its relevance to the study. The participants involved in the pretest were not included in the main data collection. The feedback gathered from the pretest helped the researchers reframe some questions and make necessary corrections to other sections, ensuring greater clarity and understanding for the main study respondents. Data collection lasted three weeks. Data was collected from respondents at their farms or community centers because it was conducive for privacy and confidentiality. Of the actual collection, 5 questionnaires could not be used in this analysis due to incomplete response or inconsistencies in the answers provided in the questionnaires. This resulted in a final sample of 211 valid responses. The sample size is deemed adequate for statistical analysis, particularly for variance-based structural equation modeling (SEM), as recommended by Hair *et al.* (2017), who suggest a minimum sample size of 100 to achieve robust and reliable SEM results.

In testing the proposed research model and hypotheses, PLS SEM was conducted. The main justification for using PLS SEM in this context is its robustness in handling complicated relationships between latent constructs and observed variables, adding to giving reliable and accurate statistical estimates (Vaithilingam *et al.*, 2024). PLS SEM was especially suitable for this study because it encompasses both factor analysis and path analysis in one, allowing the

investigation of both the measurement and structural models simultaneously (Hair *et al.*, 2020). Reliability of the internal consistency was measured by Cronbach's alpha and composite reliability (CR), which is acceptable if it is 0.70 or above (Youssef *et al.*, 2023). Convergent validity was verified by AVE, which should be greater than 0.50 for each construct (Cheung *et al.*, 2024). Factor loadings for each indicator were also checked. The recommended threshold is at least 0.50 or higher to confirm that the individual items are adequate (Baharum *et al.*, 2023). Discriminant validity was based on the Fornell–Larcker criterion, whereby the root square of the AVE for each construct should be higher than its correlations with other constructs. This can establish that the constructs indeed differ from one another (Afthanorhan *et al.*, 2021). Assessment of this model's goodness of fit also followed to establish its appropriateness. The SRMR was estimated, and the value for a good fit is < 0.08 (Hu and Bentler, 1998). The global fit indices were considered to confirm that the structure model fitted well with the data. This therefore gave the analytical process the real strength to ensure that whatever results were realized from it were reliable and meaningful.

4. Results

4.1 Demographic profiles

Table 1 summarizes the characteristics of the respondents involved in this survey. As shown in the table, 52.1% of the sample respondents were between the ages of 25–34. This was followed by participant within the age bracket of 35–44, representing 27.5% of the sample respondents. The percentages are in line with prior studies like Osei *et al.* (2022); and Osei *et al.* (2017) which indicate that young people in Ghana are now taking up farming. In terms of gender, 48.8% of the study participants were male while 51.2% were female. In terms of educational level of the participants, the majority (i.e. 49.8%) had attained a bachelor's degree, while 41.2% had attained a postgraduate degree; either Master's or PhD. The remaining participants had a secondary education or a Diploma certification representing 1.9% and 6.6%, respectively. Further, with regards to the type of farming been done by the study participants, can be seen from Table 1 that most of them were into crop farming (i.e. 48.8%), followed by mixed farming making up 25.6% of the study participants, then livestock farming which constituted of 19.9% of the sample population. The other 5.7% of the participants indicated that they into other form of farming like backyard farming or garden farming. In terms of the participants farm size, the dominant farm size used by the participants in cultivating or rearing were less than 2 acres, followed by 2–5 acres. Finally, in terms of the kind of AI-based technologies which are used by the sample respondent, the study revealed that majority of the participants, that is 40.7% uses AI-powered irrigation system in their farming practices. This was followed by AI-driven crop monitoring (25.2%), then AI-based pest and disease detection making up 16.6%. And AI-powered marketing pricing tools (11.8%) and AI for soil analysis and fertilization (5.7%) were the less AI-based technologies used by the sample population.

4.2 Assessment of measurement model

In the initial stage of the study, the reliability test and convergent validity test was carried out. In this investigation, items' loadings, average variance extracted (AVE), rho_A and composite reliability (CR) were critically analysed. The results are shown in Table 2. With reference to the mentioned table, items' loadings were more than 0.6, which fulfilled the value recommended by Hair *et al.* (2017). As for the AVE threshold, the AVE should exceed 0.5 (Ab Hamid *et al.*, 2017). In this study, the AVEs were in the range of 0.68 and 0.87, thus were acceptable. Furthermore, the CR value ranged from 0.90–0.95, again consistent with the value suggested by Hair *et al.* (2020). Also, as indicated in Table 2, the Cronbach's alpha (CA) values for each variable are above 0.70 as recommended by Hair *et al.* (2017).

Table 1. Demographic profiles

Participants demographics	Frequency	%
<i>Age</i>		
Below 25	25	11.8
25–34	110	52.1
35–44	58	27.5
45–54	16	7.6
55 and above	2	0.9
<i>Gender</i>		
Male	103	48.8
Female	108	51.2
<i>Educational level</i>		
Primary education	1	0.5
Secondary education	4	1.9
Diploma/higher national diploma	14	6.6
Bachelor's degree	105	49.8
Postgraduate (masters/PhD)	87	41.2
<i>Type of farming</i>		
Crop framing	103	48.8
Livestock framing	42	19.9
Mixed framing	54	25.6
Others	12	5.7
<i>Farm size</i>		
Less than 2 acres	127	60.2
2–5 acres	54	25.6
6–10 acres	22	10.4
More than 10 acres	8	3.8
<i>AI-based technologies used</i>		
AI-powered irrigation system	86	40.7
AI-driven crop monitoring system	53	25.2
AI-based pest and disease detection	35	16.6
AI for soil analysis and fertilization	12	5.7
AI-powered marketing pricing tools	25	11.8
Total	211	100.0

Source(s): Created by the authors, 2025

Once the test for convergent validity was finalized, the next process was to check for discriminant validity. As evident in the literature, this test had been used by [Fornell and Larcker \(1981\)](#). This study uses the Fornell–Larcker analysis for discriminant validity for reference purpose. As revealed in [Table 3](#), all constructs exhibit sufficient or satisfactory discriminant validity ([Fornell and Larcker, 1981](#)), in which the square root of AVE (diagonal) is larger than the correlations (off-diagonal) for all reflective constructs. Further, to ensure that there was no multi-collinearity among variables, the study also assessed the variance inflation factor (VIF) as shown in [Table 3](#).

4.3 Assessment of structural model

In assessing the model's robustness, we first examined the VIF. From [Table 3](#), the highest VIF value was 1.852, indicating the absence of multicollinearity. The study also recorded

Table 2. Results of measurement model

Constructs	Codes	Loadings	Cronbach's alpha	rho_A	CR	AVE
AI adoption			0.847	0.873	0.896	0.683
AI technologies improve my farming efficiency	AI_1	0.802				
I find AI-driven tools easy to use and integrate into my farming activities	AI_2	0.764				
I am willing to invest in AI technologies to enhance my agricultural productivity	AI_3	0.880				
I believe AI has a transformative role in sustainable agriculture	AI_4	0.858				
Market orientation strategies			0.926	0.927	0.953	0.871
Access to markets						
AI-based platforms improve my access to potential buyers and suppliers	AM_1	0.937				
AI helps me better connect with markets beyond my immediate locality	AM_2	0.937				
AI-driven data analytics improve my understanding of consumer demand	AM_3	0.925				
Pricing mechanisms			0.906	0.907	0.941	0.843
AI tools provide accurate price forecasts for my agricultural products	PM_1	0.89				
AI enhances my ability to set competitive yet profitable prices	PM_2	0.944				
AI-driven market insights help me reduce revenue losses due to price fluctuations	PM_3	0.919				
Environmental impact			0.900	0.900	0.938	0.833
Crop management						
AI improves my ability to monitor soil health and crop conditions	CM_1	0.911				
AI-driven predictive analytics enhance my decision-making in crop production	CM_2	0.920				
Reducing carbon footprint			0.894	0.897	0.934	0.824
AI applications contribute to reducing greenhouse gas emissions on my farm	RCF_1	0.900				
AI-driven farming practices lower dependency on chemical inputs	RCF_2	0.909				
Using AI in my farm operations reduces waste and enhances sustainability	RCF_3	0.915				
Resource efficiency			0.869	0.87	0.92	0.793
AI helps me optimize water and energy use in my farming operations	RE_1	0.892				
AI-driven insights improve my fertilizer and pesticide application efficiency	RE_2	0.924				
The use of AI reduces overall farm input wastage	RE_3	0.855				

Source(s): Created by the authors, 2025

Table 3. Discriminant validity with HTMT ratio and structural model assessment

Constructs	AI	AM	CM	PM	VIF	R ²
AI					1.852	
AM	0.726				3.386	0.432
ES	0.817	0.799				0.658
PM	0.709	0.895	0.79		3.226	0.404

Source(s): Created by the authors, 2025

an SRMR value of 0.062, thereby indicating a good model fit (Hair *et al.*, 2017). Additionally, the Stone–Geisser criterion (Q^2 value) was above 0.00. The R^2 values for the constructs were all above 0.20, thereby establishing the model’s explanatory accuracy (Sarstedt *et al.*, 2023). Results from the structural model helped to analyze the proposed hypotheses using nonparametric bootstrapping with 1,158 cases and 10,000 subsamples (Sarstedt *et al.*, 2023). From Table 4, the findings indicate that relationships between $AI \rightarrow RE$ ($\beta = 0.385$, $p = 0.00$), $AI \rightarrow CM$ ($\beta = 0.424$, $p = 0.00$) and $AI \rightarrow RCF$ ($\beta = 0.252$, $p = 0.00$) were all positively and significantly related. This indicates the confirmation of the hypothesized relationships $H1$, $H2$ and $H3$. A graphical representation based on Figure 2 helps visualize the impact of AI adoption on environmental impact, with the mediating relationship.

Table 4. Direct relationship and mediation relationship

Paths	Beta	SD.	t-stats	p-values	Decision
<i>Direct relationship</i>					
$AI \rightarrow ES$	0.391	0.080	4.909	0.000	Supported
<i>Mediation relationship</i>					
$AI \rightarrow AM \rightarrow ES$	0.184	0.064	2.856	0.004	Supported
$AI \rightarrow PM \rightarrow ES$	0.150	0.053	2.833	0.005	Supported

Source(s): Created by the authors, 2025

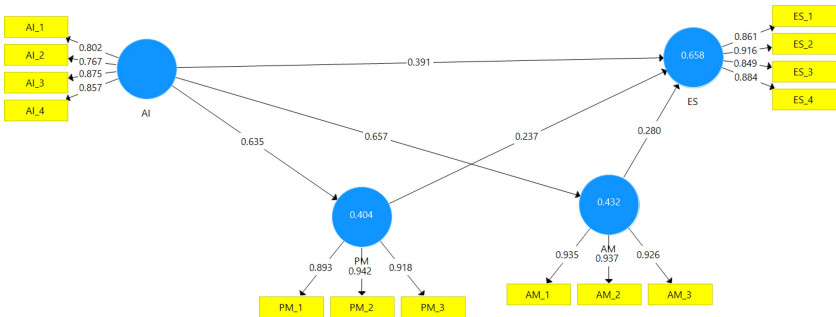


Figure 2. The structural paths
Source: Created by the authors, 2025

4.4 Mediation analysis

The study hypothesized *H4a–H4c*, which is Access to Market (AM) and *H5a–H5c*, which is Price Mechanisms (PM) could mediate the relationship between the dimensions of AI adoption (AI) and Environmental Impact dimensions (Resource Efficiency, Crop Management, Reduce Carbon Footprint). The result established that Access to Market mediated the relationship between AI adoption and reduced carbon footprint ($\beta = 0.142$, $p = 0.029$), thus confirming *H4c* of the study’s proposed hypothesis. Further, the study identified that *H4a* and *H4b* was statistically not significant (i.e. the mediating path of Access Market between Resource Efficiency and Crop Management). As such, *H4a–H4b* was not supported by the study data. The mediation results are exhibited in [Table 5](#). The study also tested the mediation relationship of Price Mechanisms between AI adoption (AI) and Resource Efficiency, Crop Management and Reduce Carbon Footprint. From the data analysis, the study was able to establish that Price Mechanisms mediated the relationship between Resource Efficiency ($\beta = 0.152$, $p = 0.025$), Crop Management ($\beta = 0.156$, $p = 0.014$) and Reduce Carbon Footprint ($\beta = 0.257$, $p = 0.000$), thus confirming *H5a–H5c* of the study’s proposed hypotheses.

5. Discussion

This study set out to investigate the role of AI-driven sustainable agriculture in reducing environmental impact, particularly focusing on how market-oriented strategies mediate this relationship. The findings of this study provide a comprehensive understanding of how Artificial Intelligence (AI) adoption significantly enhances aspects of environmental performance such as crop management (CM), resource efficiency (RE) and the reduction of carbon footprint (RCF). From the direct relationships, AI adoption was found to have a strong and statistically significant positive effect on crop management ($\beta = 0.424$, $p = 0.000$), resource efficiency ($\beta = 0.385$, $p = 0.000$) and reducing carbon footprint ($\beta = 0.252$, $p = 0.000$). These results confirm the *H1*, *H2* and *H3*, indicating that AI-driven technologies, when integrated into agricultural practices, have a direct beneficial impact on key dimensions of sustainable agriculture. This aligns with prior literature ([Adewusi et al., 2024](#); [Ahmed and Khan, 2024](#); [Padhiary et al., 2025](#)), suggesting that AI enables precision farming, optimized use of resources and better crop monitoring and forecasting, all of which contribute to improved productivity and minimized environmental degradation.

Beyond these direct relationships, the study further explored the mediating role of market-oriented strategies, particularly access to market (AM) and pricing mechanisms (PM). The mediation study provided a more detailed picture of how AI adoption affects environmental outcomes indirectly. The path $AI \rightarrow AM \rightarrow RCF$ ($\beta = 0.142$, $p = 0.029$) was statistically significant, confirming *H4c*. This implies that access to markets mediates the

Table 5. Mediation relationship

Paths	Beta	SD.	t-stats	p-values	Decision
AI → AM → RE	0.134	0.087	1.539	0.124	Not supported
AI → AM → CM	0.125	0.068	1.841	0.066	Not supported
AI → AM → RCF	0.142	0.065	2.192	0.029	Supported
AI → PM → RE	0.152	0.067	2.255	0.025	Supported
AI → PM → CM	0.156	0.063	2.473	0.014	Supported
AI → PM → RCF	0.257	0.063	4.083	0.000	Supported

Source(s): Created by the authors, 2025

relationship between AI adoption and the reduction of carbon footprint. In practical terms, this suggests that AI's contribution to reducing environmental impact is amplified when farmers have better access to markets, possibly due to reduced post-harvest losses, efficient logistics, and higher incentives for adopting sustainable practices. However, the other access-to-market mediation paths, namely, $AI \rightarrow AM \rightarrow RE$ and $AI \rightarrow AM \rightarrow CM$, were not statistically significant ($p = 0.124$ and $p = 0.066$, respectively), leading to the rejection of *H4a* and *H4b*. This implies that while market access is crucial for reducing carbon emissions, it may not play a strong mediating role in improving resource efficiency or crop management through AI. This could be because improvements in RE and CM are more technical and operational in nature and may not directly depend on market linkage.

On the other hand, pricing mechanisms emerged as a stronger and more consistent mediator. The mediation paths $AI \rightarrow PM \rightarrow RE$ ($\beta = 0.152$, $p = 0.025$), $AI \rightarrow PM \rightarrow CM$ ($\beta = 0.156$, $p = 0.014$) and $AI \rightarrow PM \rightarrow RCF$ ($\beta = 0.257$, $p = 0.000$) were all statistically significant, supporting *H5a*, *H5b* and *H5c*. This underscores the critical role of pricing strategies in enabling AI to achieve environmental benefits. Fair and transparent pricing likely encourages farmers to invest in AI technologies and sustainable practices, knowing they will receive value-aligned compensation. The findings imply that well-designed market incentives and pricing structures can drive AI adoption in ways that positively affect the environment.

6. Practical implications

The results of this study present several practical implications for policymakers, agricultural technology developers, market intermediaries and farmers. First and foremost, the evidence of AI's significant contribution to environmental sustainability offers a clear signal to agricultural policymakers that investing in and promoting AI technologies is a viable strategy for addressing climate-related challenges in the agricultural sector. Policymakers can use these findings to justify budget allocations toward digital agriculture programs, environmental conservation schemes, and sustainability incentives. Further, for agricultural extension services and development practitioners, the findings underscore the importance of integrating AI training and education into extension programs. As AI adoption has been proven to directly improve crop management and resource efficiency, extension officers must be trained not just in traditional agronomy but also in digital tools that can optimize input use, forecast crop conditions and reduce waste. Subsequently, the study's confirmation of AI's impact on sustainable farming outcomes provides strong justification for agritech companies to tailor their innovations to meet environmental objectives. Furthermore, the mediating effect of pricing mechanisms suggests that technology developers must work closely with supply chain actors and market institutions to ensure that AI-enabled farmers receive appropriate value for their outputs. Additionally, the study highlights the tangible environmental and productivity gains that farmers can achieve by embracing AI technologies. Whether through automated irrigation, drone surveillance, smart planting schedules or data-driven input management, farmers can optimize resource use, reduce environmental impact and access premium markets. These benefits, however, are best realized when they operate within a supportive ecosystem that includes affordable AI tools, reliable pricing schemes and access to competitive markets. Finally, academics and researchers in agricultural and environmental studies can use this research as a basis to further investigate sector-specific challenges in AI integration. Given the differences in significance between market access and pricing mechanisms as mediators, future research might delve into the contextual and infrastructural factors that influence these dynamics.

6.1 *Managerial implications*

The study's results provide practical implications for SME farmers, agribusiness managers, and policymakers. First, AI-driven agricultural technologies require significant investments, considering their positive impact on crop management, carbon emission reduction and improved resource efficiency. These can be supported by governmental and developmental agencies through the development of digital infrastructure, training programs and subsidies. In addition, a key strategic focus should be improving market access for farmers, particularly for its potential in minimizing carbon emissions. SME agribusinesses should consider integrating AI adoption strategies and market-based incentives. For example, leveraging contract farming and value chain collaborations can help introduce pricing models that encourage sustainability practices.

6.2 *Theoretical implications*

This study contributes to the current research by empirically confirming the importance of artificial intelligence on environmental issues. By incorporating resource optimization and, carbon footprint reduction and crop management, the research moves beyond the fragmented analyses of the past to provide a unified view on environmental outcomes. Also, the study moved beyond the “technological determinism” of AI-driven agricultural technologies (Altayeb *et al.*, 2024), this research uses a contingency-based view of technology adoption. The study argued that AI-driven agricultural technologies are not linear but are mediated by the economic environment in which farmers operate. Finally, the research provides a more context-sensitive perspective of sustainability transitions in agriculture, arguing that technological capabilities are insufficient; rather, a synergy between technology, institutional support and market conditions.

7. Conclusion

From the study findings, it was concluded that AI adoption plays a pivotal role in enhancing sustainable agricultural practices by directly improving crop management, promoting resource efficiency and contributing to the reduction of carbon footprints. These outcomes affirm that the integration of AI technologies into agricultural systems is not merely a technological trend but a transformative force capable of driving environmental sustainability. The significant direct effects of AI on crop management and resource efficiency show that AI tools such as precision agriculture systems, automated irrigation and intelligent monitoring systems are critical in optimizing agricultural processes. Equally, the positive influence on reducing carbon footprint demonstrates the potential of AI to contribute to global climate action goals by minimizing the environmental burden of farming activities. Furthermore, the study's mediation analysis establishes that market-oriented strategies serve as valuable mechanisms in amplifying the sustainability effects of AI adoption. Notably, pricing mechanisms were found to consistently mediate the relationship between AI and all three environmental dimensions, underscoring their strategic importance. Access to markets was only effective in mediating the AI-carbon footprint link, indicating that while market linkages are important, they may not always be the most effective lever for driving efficiency or crop management outcomes.

8. Direction for future studies

This study highlights the importance of AI-driven sustainable agriculture and how market-oriented strategies mediate its impact. However, it also identifies key areas for further research. Future studies should explore the qualitative aspect of access to market between resource efficiency and crop management, as it appears not to strengthen the proposed

relationships. Also, demographics of the respondents can be used as confounders on crop management. Further, contextual barriers, such as infrastructure, education and socio-economic factors, that affect AI adoption, can be considered and tested in other developing countries. The current research treats agriculture homogeneously, but variations in crops, climate and policy demand more region-specific analysis. Further, longitudinal studies are also needed to track AI's evolving effects on sustainability over time. Additionally, broader market factors like supply chain transparency, certification systems and consumer demand for sustainable products should be considered as potential influencers, future studies could expand the scope of market-based mediators.

References

- Ab Hamid, M.R., Sami, W. and Sidek, M.M. (2017), "Discriminant validity assessment: use of Fornell and Larcker criterion versus HTMT criterion", In *Journal of Physics: Conference Series*, IOP Publishing, Vol. 890 No. 1, p. 12163.
- Abdussalam, A.V. and Auladi, G.A. (2024), "Pushing boundaries: AI and computer science in the era of technological revolution", *TechComp Innovations: Journal of Computer Science and Technology*, Vol. 1 No. 1, pp. 1-9.
- Abokyi, E., Strijker, D., Asiedu, K.F. and Daams, M.N. (2020), "The impact of output price support on smallholder farmers' income: evidence from maize farmers in Ghana", *Heliyon*, Vol. 6 No. 9.
- Adewusi, A.O., Asuzu, O.F., Olorunsogo, T., Iwuanyanwu, C., Adaga, E. and Daraojimba, D.O. (2024), "AI in precision agriculture: a review of technologies for sustainable farming practices", *World Journal of Advanced Research and Reviews*, Vol. 21 No. 1, pp. 2276-2285.
- Afful-Koomson, T. and Fonta, W. (2015), *Economic and Financial Analyses of Small and Medium Food Crops Agro-Processing Firms in Ghana*, United Nations University Institute for Natural Resourc.
- Afthanorhan, A., Ghazali, P.L. and Rashid, N. (2021), "Discriminant validity: a comparison of CBSEM and consistent PLS using Fornell and Larcker and HTMT approaches", In *Journal of Physics: Conference Series*, Vol. 1874 No. 1, p. 12085.
- Aggarwal, S., Bansal, S. and Goel, R. (2024), "AI In agriculture: a looming challenge, a gleaming opportunity", *International Journal of Engineering Science and Humanities*, Vol. 14 No. Special Issue 1, pp. 43-52.
- Agholor, A.I., Ogujiuba, K.K. and Shongwe, N.I. (2023), "Determinants of small farmers access to agricultural markets in South Africa", *Agricultural Science and Technology*, Vol. 15 No. 1.
- Ahmed, S. and Khan, A. (2024), "The impact of artificial intelligence on sustainable agriculture: developing advanced computing models for environmental and economic benefits", *Journal of Advanced Computing Systems*, Vol. 4 No. 6, pp. 1-8.
- Ahuja, S. and Mehra, P. (2023), "Sustainable artificial intelligence solutions for agricultural efficiency and carbon footprint reduction in India", In *Agricultural Economics and Agri-Food Business*, IntechOpen.
- Aidoo, M., Prah, S., Asante, I.S., Sackey, C.K. and Asante, B.O. (2025), "Cultivating a thriving agricultural sector: unveiling the drivers of farmer participation in agricultural development interventions in Ghana", *Social Sciences and Humanities Open*, Vol. 11, p. 101297.
- Al-Atrash, Y.E., Wishah, A.T., Abul-Omreen, T.H. and Abu-Naser, S.S. (2020), "Modeling cognitive development of the balance scale task using ANN", *International Journal of Academic Information Systems Research (IJAIRS)*, Vol. 4 No. 9.

- Altayeb, J.M., Eleyan, H., Wishah, N.D., Elmahmoum, A.E., Khalil, A.J., Abu-Nasser, B.S. and Abu-Naser, S.S. (2024), *AI-Driven Innovations in Agriculture: Transforming Farming Practices and Outcomes*, IJAAR.
- Amadasun, D.O. and Mutezo, A.T. (2022), "Effect of market-driven strategies on the competitive growth of SMEs in Lesotho", *Journal of Innovation and Entrepreneurship*, Vol. 11 No. 1, p. 21.
- Ayeni, O.O., Al Hamad, N.M., Chisom, O.N., Osawaru, B. and Adewusi, O.E. (2024), "AI in education: a review of personalized learning and educational technology", *GSC Advanced Research and Reviews*, Vol. 18 No. 2, pp. 261-271.
- Azumah, F.D., Onzaberigu, N.J. and Adongo, A.A. (2023), "Gender, agriculture and sustainable livelihood among rural farmers in Northern Ghana", *Economic Change and Restructuring*, Vol. 56 No. 5, pp. 3257-3279.
- Baharum, H., Ismail, A., Awang, Z., McKenna, L., Ibrahim, R., Mohamed, Z. and Hassan, N.H. (2023), "The study adapted instruments based on confirmatory factor analysis (CFA) to validate measurement models of latent constructs", *International Journal of Environmental Research and Public Health*, Vol. 20 No. 4, p. 2860.
- Balkrishna, A., Pathak, R., Kumar, S., Arya, V. and Singh, S.K. (2023), "Smart agricultural technology", *Precision Agriculture (PA)(2030)*, p. 5.
- Borah, S.K., Pal, D., Sarkar, S. and Sethi, L.N. (2025), "AI-Powered drones for sustainable agriculture and precision farming", In *Advancing Global Food Security With Agriculture 4.0 and 5.0*, IGI Global Scientific Publishing, pp. 69-98.
- Cheung, G.W., Cooper-Thomas, H.D., Lau, R.S. and Wang, L.C. (2024), "Reporting reliability, convergent and discriminant validity with structural equation modeling: a review and best-practice recommendations", *Asia Pacific Journal of Management*, Vol. 41 No. 2, pp. 745-783.
- Corsi, A., Novelli, S. and Pettenati, G. (2018), "Producer and farm characteristics, type of product, location: determinants of on-farm and off-farm direct sales by farmers", *Agribusiness*, Vol. 34 No. 3, pp. 631-649.
- Dawn, N., Ghosh, T., Ghosh, S., Saha, A., Mukherjee, P., Sarkar, S., Guha, S. and Sanyal, T. (2023), "Implementation of artificial intelligence, machine learning, and internet of things (IoT) in revolutionizing agriculture: a review on recent trends and challenges", *International Journal of Experimental Research and Review*, Vol. 30.
- De Toni, D., Reche, R.A. and Milan, G.S. (2022), "Effects of market orientation, innovation strategies and value-based pricing on market performance", *International Journal of Productivity and Performance Management*, Vol. 71 No. 8, pp. 3556-3580.
- Eli-Chukwu, N.C. (2019), "Applications of artificial intelligence in agriculture: a review", *Engineering, Technology and Applied Science Research*, Vol. 9 No. 4.
- Elkington, J. (1994), "Towards the sustainable corporation: win-win-win business strategies for sustainable development", *California Management Review*, Vol. 36 No. 2, pp. 90-100.
- Elkington, J. (1997), "The triple bottom line", *Environmental Management: Readings and Cases*, Vol. 2 No. 1997, pp. 49-66.
- Elshaikh-Hayaty, A., Elsheikh, N.E. and Mabrouki, N.J. (2024), "Applications of artificial intelligence in precision irrigation", *Journal of Environmental and Earth Sciences*, Vol. 6 No. 2, pp. 176-186.
- Essel, B.K.C., Donkor, E., Ullah, A., Adams, F. and Bavorova, M. (2026), "Youth entrepreneurship, credit access and post-disaster business recovery: evidence from informal food enterprises in Ghana", *Journal of International Development*, Vol. 38 No. 3.
- Fasciolo, B., Panza, L. and Lombardi, F. (2024), "Exploring the integration of industry 4.0 technologies in agriculture: a", *Comprehensive Bibliometric Review. Sustainability*, Vol. 16 No. 20, p. 8948.

-
- Fornell, C. and Larcker, D.F. (1981), "Evaluating structural equation models with unobservable variables and measurement error", *Journal of Marketing Research*, Vol. 18 No. 1, pp. 39-50.
- Gebremedhin, B. (2010), "Commercialization of smallholders: does market orientation translate into market participation?", *ILRI (Aka ILCA and ILRAD)*, Vol. 22.
- Ghosh, M. and Thirugnanam, A. (2021), "Introduction to artificial intelligence", In *Artificial Intelligence for Information Management: A Healthcare Perspective*, Springer Singapore, Singapore, pp. 23-44.
- Gu, W., Wang, J., Hua, X. and Liu, Z. (2021), "Entrepreneurship and high-quality economic development: based on the triple bottom line of sustainable development", *International Entrepreneurship and Management Journal*, Vol. 17 No. 1, pp. 1-27.
- Hair, J.F., Jr, Howard, M.C. and Nitzl, C. (2020), "Assessing measurement model quality in PLS-SEM using confirmatory composite analysis", *Journal of Business Research*, Vol. 109, pp. 101-110.
- Hair, J.F., Jr, Matthews, L.M., Matthews, R.L. and Sarstedt, M. (2017), "PLS-SEM or CB-SEM: updated guidelines on which method to use", *International Journal of Multivariate Data Analysis*, Vol. 1 No. 2, pp. 107-123.
- Hamed, M.A., El-Habib, M.F., Sababa, R.Z., Al-Hanjor, M.M., Abunasser, B.S. and Abu-Naser, S.S. (2024), "Artificial intelligence in agriculture: enhancing productivity and sustainability".
- Herath, H.M.K.K.M.B. and Mittal, M. (2022), "Adoption of artificial intelligence in smart cities: a comprehensive review", *International Journal of Information Management Data Insights*, Vol. 2 No. 1, p. 100076.
- Holzinger, A., Langs, G., Denk, H., Zatloukal, K. and Müller, H. (2019), "Causability and explainability of artificial intelligence in medicine", *Wiley Interdisciplinary Reviews. Data Mining and Knowledge Discovery*, Vol. 9 No. 4, p. e1312.
- Hoseinzadeh, S. and Garcia, D.A. (2024), "Ai-driven innovations in greenhouse agriculture: reanalysis of sustainability and energy efficiency impacts", *Energy Conversion and Management*, Vol. X, p. 100701.
- Hu, L.T. and Bentler, P.M. (1998), "Fit indices in covariance structure modeling: sensitivity to under parameterized model misspecification", *Psychological Methods*, Vol. 3 No. 4, p. 424.
- Hu, T., Zhang, X., Khanal, S., Wilson, R., Leng, G., Toman, E.M., ... Zhao, K. (2024), "Climate change impacts on crop yields: a review of empirical findings, statistical crop models, and machine learning methods", *Environmental Modelling and Software*, Vol. 179.
- Ilak, P., Herenčić, L., Rajšl, I., Raos, S. and Tomšić, Ž. (2021), "Equilibrium pricing with Duality-Based method: approach for Market-Oriented capacity remuneration mechanism", *Energies*, Vol. 14 No. 3, p. 567.
- Issa, H., Jabbouri, R. and Palmer, M. (2022), "An artificial intelligence (AI)-readiness and adoption framework for AgriTech firms", *Technological Forecasting and Social Change*, Vol. 182, p. 121874.
- Javaid, M., Haleem, A., Khan, I.H. and Suman, R. (2023), "Understanding the potential applications of artificial intelligence in agriculture sector", *Advanced Agrochem*, Vol. 2 No. 1, pp. 15-30.
- Jha, K., Doshi, A., Patel, P. and Shah, M. (2019), "A comprehensive review on automation in agriculture using artificial intelligence", *Artificial Intelligence in Agriculture*, Vol. 2, pp. 1-12.
- Jha, R. and Sharma, A. (2025), "Adoption of agricultural resources for sustainable environment", In *Building Inclusive Global Knowledge Societies for Sustainable Development*, IGI Global Scientific Publishing, pp. 29-42.
-

- Kamilaris, A. and Prenafeta-Boldú, F.X. (2018), "Deep learning in agriculture: a survey", *Computers and Electronics in Agriculture*, Vol. 147, pp. 70-90.
- Kanaan-Jebna, A., Baharudi, A.S. and Alabdullah, T.T.Y. (2022), "Entrepreneurial orientation, market orientation, managerial accounting and manufacturing SMEs satisfaction", *Journal of Accounting Science*, Vol. 6 No. 1, pp. 1-14.
- Keung, E.Z., McElroy, L.M., Ladner, D.P. and Grubbs, E.G. (2020), "Defining the study cohort: inclusion and exclusion criteria", In *Clinical Trials*, Springer International Publishing, Cham, pp. 47-58.
- Kohli, A.K. and Jaworski, B.J. (1990), "Market orientation: the construct, research propositions, and managerial implications", *Journal of Marketing*, Vol. 54 No. 2, pp. 1-18.
- Koryo-Dabrah, A., Ansong, R.S., Setorglo, J. and Steiner-Asiedu, M. (2021), "Food and nutrition security situation in Ghana: nutrition implications for national development. *African Journal of Food*", *Agriculture, Nutrition and Development*, Vol. 21 No. 5, pp. 18005-18018.
- Kumar, L. (2024), "The role of agriculture in economic development", *Journal of Environmental Agriculture and Agroecosystem Management*, Vol. 1 No. 1, pp. 62-68.
- Kumari, K., Mirzakhani Nafchi, A., Mirzaee, S. and Abdalla, A. (2025), "AI-driven future farming: achieving climate-smart and sustainable agriculture", *AgriEngineering*, Vol. 7 No. 3, p. 89.
- Li, H., Lu, Z., Zhang, Z. and Tanasescu, C. (2025), "How does artificial intelligence affect manufacturing firms' energy intensity?", *Energy Economics*, Vol. 141, p. 108109.
- McKinsey & Company (2019), "Global AI survey: AI proves its worth, but few scale impact", *McKinsey Analytics*, available at: www.mckinsey.com/featured-insights/artificial-intelligence/global-ai-survey-ai-proves-its-worth-but-few-scale-impact
- Magesa, M.M., Michael, K. and Ko, J. (2020), "Access and use of agricultural market information by smallholder farmers: measuring informational capabilities", *The Electronic Journal of Information Systems in Developing Countries*, Vol. 86 No. 6, p. e12134.
- Mahmoud, M.A., Blankson, C., Owusu-Frimpong, N., Nwankwo, S. and Trang, T.P. (2016), "Market orientation, learning orientation and business performance: the mediating role of innovation", *International Journal of Bank Marketing*, Vol. 34 No. 5, pp. 623-648.
- Mahmoud, M.A., Hinson, R.E. and Duut, D.M. (2019), "Market orientation and customer satisfaction: the role of service quality and innovation", *International Journal of Business and Emerging Markets*, Vol. 11 No. 2, pp. 144-167.
- Meemken, E.M. and Bellemare, M.F. (2020), "Smallholder farmers and contract farming in developing countries", *Proceedings of the National Academy of Sciences*, Vol. 117 No. 1, pp. 259-264.
- Muchhadiya, R.M., Gohil, B., Yadahalli, V., Hm, A.U.R., Siddiqua, A., Khayum, A., ... Kumar, S. (2024), "Feeding the world: agronomic innovations to meet the challenges of a growing population", *International Journal of Research in Agronomy*, Vol. 7 No. 7, pp. 790-802.
- Muhie, S.H. (2022), "Novel approaches and practices to sustainable agriculture", *Journal of Agriculture and Food Research*, Vol. 10, p. 100446.
- Naruetharadhol, P., Ketkaew, C. and Srisathan, W.A. (2022), "Innovative price-setting approaches to high-value products: a pricing method for agribusiness farmers", *Heliyon*, Vol. 8 No. 9.
- Narver, J.C. and Slater, S.F. (1990), "The effect of a market orientation on business profitability", *Journal of Marketing*, Vol. 54 No. 4, pp. 20-35.
- Nindi, T., Ricker-Gilbert, J. and Bauchet, J. (2024), "Incentive mechanisms to exploit intraseasonal price arbitrage opportunities for smallholder farmers: experimental evidence from Malawi", *American Journal of Agricultural Economics*, Vol. 106 No. 1, pp. 330-353.
- OECD (2016), *OECD Science, Technology and Innovation Outlook*, OECD, Paris.

- Omia, E., Bae, H., Park, E., Kim, M.S., Baek, I., Kabenge, I. and Cho, B.K. (2023), "Remote sensing in field crop monitoring: a comprehensive review of sensor systems, data analyses and recent advances", *Remote Sensing*, Vol. 15 No. 2, p. 354.
- Osei, R., Adusei-Antwi, J.R., Gyabaah, V. and Yeboah, R.A. (2022), "Assessing youth perceptions and knowledge of modern agriculture: a case of youth in Tano North municipality of Ahafo region", *International Journal Of All Research Writings*, Vol. 5 No. 7, pp. 11-20.
- Osei, S.K., Folitse, B.Y., Dzandu, L.P. and Obeng-Koranteng, G. (2017), "Sources of information for urban vegetable farmers in Accra, Ghana", *Information Development*, Vol. 33 No. 1, pp. 72-79.
- Otekunrin, O.A., Momoh, S. and Ayinde, I.A. (2019), "Smallholder farmers' market participation: concepts and methodological approach from Sub-Saharan Africa", *Current Agriculture Research Journal*, Vol. 7 No. 2, p. 139.
- Padhiary, M., Hoque, A., Prasad, G., Kumar, K. and Sahu, B. (2025), "Precision agriculture and AI-Driven resource optimization for sustainable land and resource management", In *Smart Water Technology for Sustainable Management in Modern Cities*, IGI Global Scientific Publishing, pp. 197-232.
- Paudel, G.P., Kc, D.B., Khanal, N.P., Justice, S.E. and McDonald, A.J. (2019), "Smallholder farmers' willingness to pay for scale-appropriate farm mechanization: evidence from the mid-hills of Nepal", *Technology in Society*, Vol. 59, p. 101196.
- Porter, J.R., Challinor, A.J., Henriksen, C.B., Howden, S.M., Martre, P. and Smith, P. (2019), "Invited review: intergovernmental panel on climate change, agriculture, and food—a case of shifting cultivation and history", *Global Change Biology*, Vol. 25 No. 8, pp. 2518-2529.
- Pretty, J. (2018), "Intensification for redesigned and sustainable agricultural systems", *Science*, Vol. 362 No. 6417, p. eaav0294.
- Puttegowda, M. and Nagaraju, S.B. (2025), "Artificial intelligence and machine learning in mechanical engineering: current trends and future prospects", *Engineering Applications of Artificial Intelligence*, Vol. 142, p. 109910.
- Quaye, W., Akon-Yamga, G., Akuffobe-Essilfie, M. and Onumah, J.A. (2024), "Technology adoption, competitiveness and new market access among SMEs in Ghana: what are the limiting factors?", *African Journal of Science, Technology, Innovation and Development*, Vol. 16 No. 7, pp. 1023-1037.
- Raihan, A.A. (2024), "Review of artificial intelligence (AI) for sustainable agriculture", In *Proceedings of the International Conference on Digitalization and Artificial Intelligence in Agricultural Management*.
- Rajak, P., Ganguly, A., Adhikary, S., *et al* (2023), "Internet of things and smart sensors in agriculture: scopes and challenges", *Journal of Agriculture and Food Research*, Vol. 14 No. 14, p. 100776.
- Ren, L., Dong, J., Liu, S., Zhang, L. and Wang, L. (2024), "Embodied intelligence toward future smart manufacturing in the era of AI foundation model", *IEEE/ASME Transactions on Mechatronics*, Vol. 30 No. 4.
- Ryan, M., Isakhanyan, G. and Tekinerdogan, B. (2023), "An interdisciplinary approach to artificial intelligence in agriculture", *NJAS: Impact in Agricultural and Life Sciences*, Vol. 95 No. 1, p. 2168568.
- Saran, C. (2019), "AI can help to reduce carbon footprint and manage environmental issues", Retrieved from: www.computerweekly.com/news/252461751/AI-can-help-to-reduce-carbon-footprint-and-manage-environmental-issues
- Sarstedt, M., Hair, J.F., Jr and Ringle, C.M. (2023), "PLS-SEM: indeed a silver bullet"—retrospective observations and recent advances", *Journal of Marketing Theory and Practice*, Vol. 31 No. 3, pp. 261-275.

- Sekhar, M., Rastogi, M., Rajesh, C.M., Saikanth, D.R.K., Rout, S., Kumar, S. and Patel, A.K. (2024), "Exploring traditional agricultural techniques integrated with modern farming for a sustainable future: a review", *Journal of Scientific Research and Reports*, Vol. 30 No. 3, pp. 185-198.
- Singh, A. (2025), "Future of jobs prospect through artificial intelligence by 2030", *Available at SSRN 5213331*.
- Singh, T., Bhadwaj, H., Verma, L., Navadia, N.R., Singh, D., Sakalle, A. and Bhardwaj, A. (2022), "Applications of AI in agriculture", *Challenges and Opportunities for Deep Learning Applications in Industry*, Vol. 4, p. 181.
- Sinha, P. and Gond, S. (2024), "The usage of cloud computing and iot in artificial intelligence", In *Fusion of Artificial Intelligence and Machine Learning in Advanced Image Processing*, Apple Academic Press, pp. 107-123.
- Sinwar, D., Dhaka, V.S., Sharma, M.K. and Rani, G. (2019), "AI-based yield prediction and smart irrigation", in *Internet of Things and Analytics for Agriculture*, Springer Singapore, Singapore, Vol. 2, pp. 155-180.
- Sutton, R.T., Pincock, D., Baumgart, D.C., Sadowski, D.C., Fedorak, R.N. and Kroeker, K.I. (2020), "An overview of clinical decision support systems: benefits, risks, and strategies for success", *NPJ digital medicine*, Vol. 3 No. 1, p. 17.
- Szymański, J., Nurzyńska, K. and Weichbroth, P. (2025), "AI-Driven sustainability in agriculture and farming", In *Digital Sustainability*, CRC Press, pp. 213-226.
- Thomas, F.B. (2022), "The role of purposive sampling technique as a tool for informal choices in a social sciences in research methods", *Just Agriculture*, Vol. 2 No. 5, pp. 1-8.
- Tilavova, M., Safarova, N., Alimova, M. and Bo'ronova, N. (2024), "Ecotourism as a sustainable development strategy: exploring the role of natural resource management", In *E3S Web of Conferences*, (Vol. 587). EDP Sciences, p. 5020.
- Tsetse, E.K.K., Mahmoud, M.A., Blankson, C. and Odoom, R. (2022), "The impact of stakeholder market orientation on sustainability performance at tourism destinations", *Management Research Review*, Vol. 45 No. 7, pp. 929-955.
- UNCTAD (2017), "Information economy report", available at: <https://unctd.org/en/pages/publication-webflyer.aspx?publicationid=1872>
- Vaithilingam, S., Ong, C.S., Moisescu, O.I. and Nair, M.S. (2024), "Robustness checks in PLS-SEM: a review of recent practices and recommendations for future applications in business research", *Journal of Business Research*, Vol. 173, p. 114465.
- Waly, M.F. (2024), "Artificial intelligence and scientific research", *Sustainability Education Globe*, Vol. 2 No. 1, pp. 1-14.
- Wang, J., Huo, Y., Guo, X. and Xu, Y. (2022), "The pricing strategy of the agricultural product supply chain with farmer cooperatives as the core enterprise", *Agriculture*, Vol. 12 No. 5, p. 732.
- World Bank (2023), "The World Bank in Ghana: overview", available at: www.worldbank.org/en/country/ghana/overview (accessed 4 August 2026).
- Xu, A., Baysari, M.T., Stocker, S.L., Leow, L.J., Day, R.O. and Carland, J.E. (2020), "Researchers' views on, and experiences with, the requirement to obtain informed consent in research involving human participants: a qualitative study", *BMC Medical Ethics*, Vol. 21 No. 1, p. 93.
- Yang, Z. and Solangi, Y.A. (2024), "Analyzing the relationship between natural resource management, environmental protection, and agricultural economics for sustainable development in China", *Journal of Cleaner Production*, Vol. 450, p. 141862.
- Youssef, N., Saleeb, M., Gebreal, A. and Ghazy, R.M. (2023), "The internal reliability and construct validity of the evidence-based practice questionnaire (EBPQ): evidence from healthcare professionals in the Eastern Mediterranean region", In *Healthcare*, Vol. 11 No. 15, p. 2168.
- Zhang, J., Mishra, A.K. and Hirsch, S. (2021), "Market-oriented agriculture and farm performance: evidence from rural China", *Food Policy*, Vol. 100, p. 102023.

Further reading

- Ali, E.B., Agyekum, E.B. and Adadi, P. (2021), "Agriculture for sustainable development: a SWOT-AHP assessment of Ghana's planting for food and jobs initiative", *Sustainability*, Vol. 13 No. 2, p. 628.
- Qamar, A. (2024), "AI-Powered solutions in climate change mitigation and environmental sustainability", *Research Corridor Journal of Engineering Science*, Vol. 1 No. 2, pp. 194-209.
- Tang, Q., Bennett, S.J., Xu, Y. and Li, Y. (2013), "Agricultural practices and sustainable livelihoods: rural transformation within the loess Plateau, China", *Applied Geography*, Vol. 41, pp. 15-23.
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